Problem Set 2

Applied Stats/Quant Methods 1

Due: October 22, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 22, 2026. No late assignments will be accepted.

Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.[1] As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, "We can solve this the easy way" to draw a bribe). The table below shows the resulting data.

[1] Fried, Lagunes, and Venkataramani (2010). "Corruption and Inequality at the Crossroad: A Multimethod Study of Bribery and Discrimination in Latin America. *Latin American Research Review.* 45 (1): 76-97.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

(a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in R).

```
# (a)
# create table
df <- matrix(data=c(14, 6, 7, 7, 7, 1), ncol = 3, byrow = TRUE)
colnames(df) <- c('Not Stopped', 'Bribe requested', 'Stopped/given
    warning')
rownames(df) <- c('Upper Class', 'Lower Class')

# make my own
df_margin <- addmargins(df)
chi_test <- 0
for (c in 1:(ncol(df_margin)-1)) {
  for (r in 1:(nrow(df_margin)-1)) {
    #print(df_margin[r,c])
    # chi_test <- chi_test + (df_margin[r,c] - (df_margin[nrow(df_margin)
    ,c]/df_margin[nrow(df_margin),ncol(df_margin)] * df_margin[r,ncol(df_
    margin)]))^2 / (df_margin[nrow(df_margin),c]/df_margin[nrow(df_margin)
    ,ncol(df_margin)] * df_margin[r,ncol(df_margin)])
    observed <- df_margin[r,c]
    row_total <- df_margin[r,ncol(df_margin)]
    col_total <- df_margin[nrow(df_margin),c]
    total <- df_margin[nrow(df_margin),ncol(df_margin)]
    expected <- (row_total*col_total)/total

    chi_test <- chi_test + ((observed-expected)^2)/expected
    }
```

```
}
chi_test
# Chi-square = 3.791168

# use build in function to check
X <- chisq.test(df)
```

The χ^2 test statistic is 3.791168.

(b) Now calculate the p-value from the test statistic you just created (in R).[2] What do you conclude if $\alpha = 0.1$?

```
# (b)
P <- pchisq(chi_test, df=(2-1)*(3-1), lower.tail = FALSE)
P1 <- X$p.value
P1
# p-value = 0.1502306
```

H_0: Driver's class and the police interaction are independent.

H_a: Driver's class and the police interaction are dependent.

Since p-value $> \alpha$, we fail to reject H_0, there is not sufficient evidence to conclude that the drivers' class and the police interaction are dependent.

[2]Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

(c) Calculate the standardized residuals for each cell and put them in the table below.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.3220306	-1.641957	1.523026
Lower class	-0.3220306	1.641957	-1.523026

```r
# (c)
find_std_residual <- function(df) {
  #out <- 0
  std_residual <- matrix(, nrow = nrow(df), ncol = ncol(df))
  df_margin <- addmargins(df)
  for (c in 1:(ncol(df_margin)-1)) {
    for (r in 1:(nrow(df_margin)-1)) {
      #print(df_margin[r,c])
      observed <- df_margin[r,c]
      row_total <- df_margin[r,ncol(df_margin)]
      col_total <- df_margin[nrow(df_margin),c]
      total <- df_margin[nrow(df_margin),ncol(df_margin)]

      expected <- (row_total*col_total)/total
      row_prop <- row_total/total
      col_prop <- col_total/total

      z <- (observed-expected)/sqrt(expected*(1-row_prop)*(1-col_prop))
      #print(observed)
      #print(row_total)
      #print(col_total)
      #print(total)
      #print(expected)
      #print(z)
      #out <- out + ((observed-expected)^2)/ expected
      print(z)
      std_residual[r,c] <- z
    }
  }
  return(std_residual)
}

std_residual <- find_std_residual(df)
```

(d) How might the standardized residuals help you interpret the results?

The standardized residuals for "Not Stopped" are relatively small, meaning that the observed frequencies are close to the expected frequencies. Therefore, these cells contribute relatively little to the chi-square statistic.

However, "Bribe Requested" and "Stopped/given warning" have relatively larger standardized residuals, meaning there are bigger differences between observed and expected frequencies. These cells contribute more to the chi-square statistic.

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.[3] Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: `https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv`

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure 1 below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

Figure 1: Names and description of variables from Chattopadhyay and Duflo (2004).

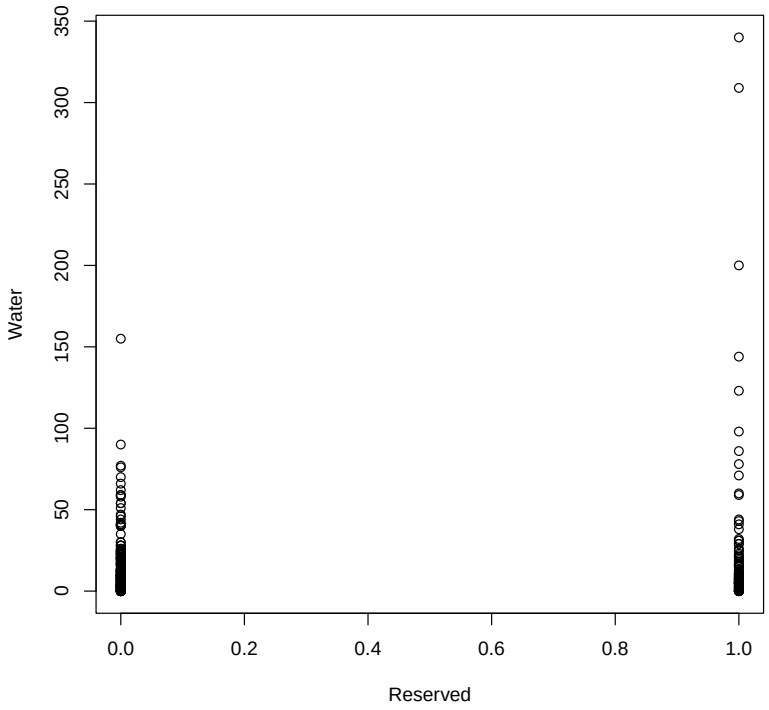
[3]Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica*. 72 (5), 1409-1443.

(a) State a null and alternative (two-tailed) hypothesis.

H_0: The reservation policy has no effect on the number of new or repaired drinking water facilities in the villages. $\rho = 0$

H_a: The reservation policy has some effects on the number of new or repaired drinking water facilities in the villages. $\rho \neq 0$

(b) Run a bivariate regression to test this hypothesis in `R` (include your code!).



```r
#(b)
df_2 <- read.csv("https://raw.githubusercontent.com/kosukeimai/qss/master
    /PREDICTION/women.csv", header=T)
summary(df_2)

pdf("reserved_water.pdf")
plot(df_2$reserved, df_2$water,xlab = "Reserved", ylab = "Water")
dev.off()

reg <- cor.test(df_2$reserved, df_2$water)
reg$p.value
# pvalue = 0.01970398
```

p-value = 0.01970398

If we have significant value α set to be 0.05. P-value $< \alpha$. We can reject the null hypotheses. We have sufficient evidence to conclude that the reservation policy has some effects on the number of the new or repaired drinking water facilities in the villages.

(c) Interpret the coefficient estimate for reservation policy.

```
#(c)
lm(water~reserved , data = df_2)
```

Intercept = 14.738 and Slope = 9.252

When the number of reservation policy number is 0, the predicted number of new or repaired drinking water facilities in the villages is 14.738, on average. When the reservation policy is applied, the predicted number of new or repaired drinking water facilities in the villages increase by 9.252, on average.